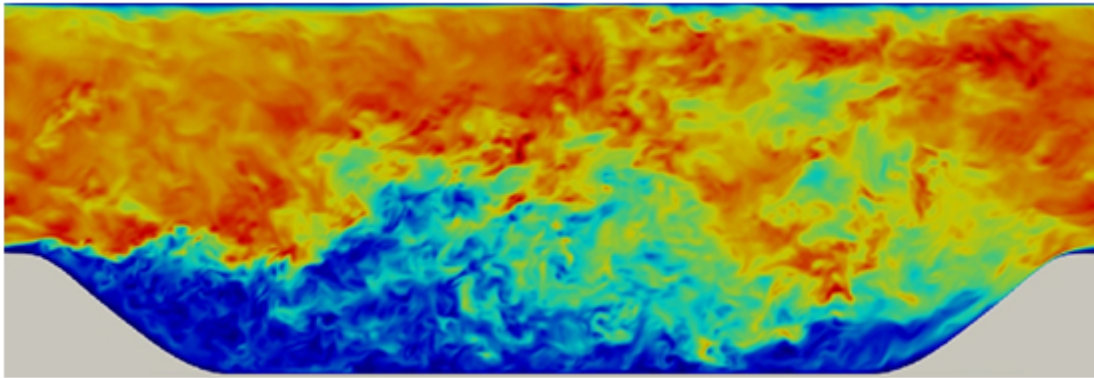


Flow Over Periodic Hills

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1. Introduction

Turbulent flows involving separation from curved surfaces occur in numerous engineering and environmental applications, including flow over terrain, vehicles, turbine components, and internal channel constrictions. Their prediction is challenging because the separation position varies spatially and temporally, while the resulting shear layer contains energetic structures that interact with the recirculation region and downstream reattachment. Flow over periodic hills has consequently become an established benchmark for evaluating turbulence models and numerical methods. The configuration combines a relatively simple and reproducible setup with pressure-induced separation, recirculation, shear-layer development, and reattachment on the lower wall [1, 2, 3]. Streamwise periodicity avoids the need to prescribe synthetic turbulent inflow conditions, while highly resolved large eddy simulation (LES), direct numerical simulation (DNS), and experimental data are available at the hill Reynolds number $Re_H = 10\,595$ for model validation.

LES resolves the larger, flow-dependent turbulent scales while modelling the smaller scales through a subgrid-scale model. This provides information about instantaneous turbulent behaviour that is unavailable from a steady Reynolds-averaged Navier–Stokes solution. However, LES accuracy depends strongly on the spatial and temporal resolution. Near-wall spacing, integral turbulent length scales, subgrid-scale activity, time-step size, and statistical sampling duration must therefore be assessed. Two-point correlations are particularly useful for determining the spatial extent of the resolved structures and the number of computational cells used to represent them [4, 5].

In this study, the periodic-hill geometry is additionally interpreted as an idealised valley exposed to a prevailing wind. A suitable location for a tourist village should experience relatively low wind loading and limited temporal and directional variability. The skin-friction coefficient and time-averaged wall shear stress (TAWSS) describe the mean near-ground loading, but do not fully represent changes in the instantaneous shear direction. Transverse wall shear stress (transWSS) accounts for wall-shear-stress components acting perpendicular to the dominant mean direction and can therefore identify multidirectional near-wall disturbances [6]. Wind-rose plots are additionally used to describe the directional distribution of the instantaneous wall shear stress.

The aim of this study is to verify and validate a WALE-based LES of turbulent flow over periodic hills at $Re_H = 10\,595$, and subsequently use the resolved near-wall characteristics to identify a region of the valley with relatively low and predictable wind conditions. A precursor k – ω SST simulation is used to characterise the flow and determine the required LES mesh resolution. The LES is verified using spatial- and temporal-resolution measures, subgrid-scale viscosity, two-point correlations, turbulence spectra, and statistical-convergence analyses. Mean velocity, Reynolds stresses, turbulent kinetic energy, skin friction, separation, and reattachment are then compared with DNS and highly resolved LES reference data. Finally, the aerodynamic environment is assessed using the skin-friction coefficient, TAWSS, transWSS, and wall-shear-stress wind-rose distributions.

2. Method

Computational configuration and numerical setup

The flow was simulated using the periodic-hill configuration established by Fröhlich et al. [1]. The hill height, $H = 1$ m, was used as the characteristic length scale. The domain extended $9H$ in the streamwise direction and $4.5H$ in the spanwise direction.

The upper wall was located at $y/H = 3.035$, giving a clearance of $2.035H$ above the hill crest at $y/H = 1$. The coordinates x , y and z denote the streamwise, wall-normal and spanwise directions, respectively, with $x/H = 0$ located at the hill crest. The computational configuration and mesh are presented in Fig. 1.

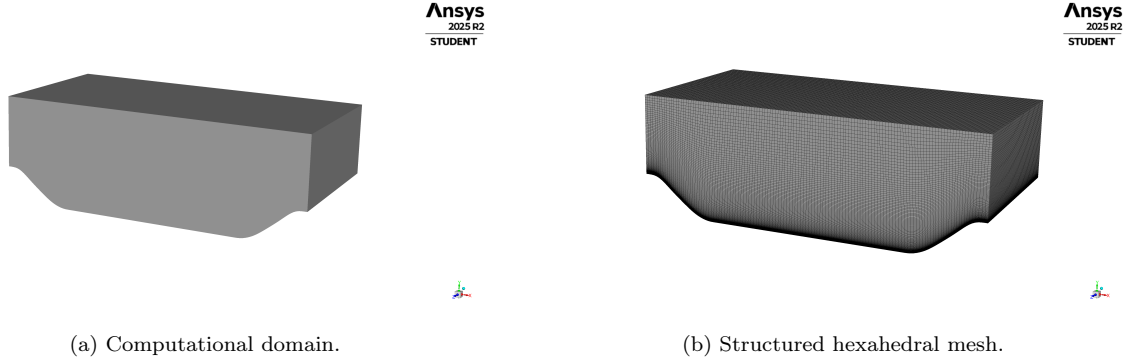


Figure 1: Periodic-hill configuration used in this study: (a) computational geometry and (b) final mesh used for both the precursor RANS and LES calculations.

The Reynolds number (Re_H) was 10 595. The bulk velocity was not imposed directly. Instead, a constant mass flow rate of $\dot{m} = 1.79 \text{ kg s}^{-1}$ was prescribed through the streamwise periodic boundary condition, corresponding to $U_b \simeq 1.00 \text{ m s}^{-1}$. An initial streamwise pressure gradient of $-2.262 \times 10^{-3} \text{ Pa m}^{-1}$ was specified. Translational periodicity was imposed on the two spanwise boundaries, while the upper wall and complete lower hill surface were treated as stationary no-slip walls. Gravity was neglected.

A fully structured hexahedral mesh was generated in ANSYS ICEM CFD using 129, 140 and 56 grid locations in the streamwise, wall-normal and spanwise directions, respectively. This corresponded to approximately 0.98 million cells. The streamwise and spanwise distributions were uniform, while the cells were clustered towards both walls using a first wall-normal cell height of $5.0 \times 10^{-4} \text{ m}$ and a growth ratio of 1.0364. The Fluent mesh-quality assessment gave a minimum orthogonal quality of 0.534 and a maximum cell aspect ratio of approximately 262. The orthogonal quality was therefore satisfactory throughout the domain. Although the maximum aspect ratio was relatively high, it was associated with the stretched structured cells required for near-wall resolution. The wall-aligned cell orientation and low growth ratio provided a smooth variation in cell dimensions. The mesh was therefore retained based on the combined orthogonal-quality, wall-unit and Courant-number assessments.

A steady precursor simulation was performed on the same mesh using the $k-\omega$ SST model. The solution was hybrid initialized and advanced for 500 iterations. Pressure-velocity coupling was treated using the Coupled algorithm, with least-squares cell-based gradients, second-order pressure interpolation, second-order-upwind momentum discretisation, and first-order-upwind discretisation for the k and ω transport equations. Convergence was assessed using a residual criterion of 10^{-5} together with the global mass-flow imbalance. The precursor RANS solution was used to estimate the mesh and time-step requirements, while the LES itself was hybrid initialized independently. Although this differed from initializing the LES directly from the precursor RANS field, approximately five flow-through times were discarded before statistical sampling was activated, reducing the influence of the initial condition on the reported statistics.

The LES was performed in ANSYS Fluent using the pressure-based transient solver and the WALE subgrid-scale model. Pressure-velocity coupling was treated using SIMPLEC. Gradients

Parameter	Value or method	Parameter	Value or method
Domain extent	$9H \times 3.035H \times 4.5H$	Minimum orthogonal quality	0.534
Reynolds number, Re_H	10 595	Maximum aspect ratio	262
Density, ρ	0.195 kg m^{-3}	Precursor model	$k-\omega$ SST
Dynamic viscosity, μ	$1.84 \times 10^{-5} \text{ Pa s}$	LES subgrid model	WALE
Mass flow rate	1.79 kg s^{-1}	LES time step	0.005 s
Grid locations	$129 \times 140 \times 56$	Iterations per time step	Maximum of 2
Approximate cell count	0.98 million	Reported simulation duration	125 s
First wall-normal spacing	$5.0 \times 10^{-4} \text{ m}$	Start of statistical sampling	45 s
Wall-normal growth ratio	1.0364	Sampling duration	80 s

Table 1: Principal physical and numerical parameters.

were evaluated using the least-squares cell-based method, pressure was discretised using the standard scheme, and momentum was discretised using bounded central differencing. Time advancement employed the bounded second-order implicit scheme, with a maximum of two solver iterations per time step. The principal setup parameters are summarised in Table 1.

The preliminary time-step estimate was obtained from the maximum precursor velocity, $U_{\max} = 1.17646 \text{ m s}^{-1}$, and the minimum effective cell dimension, $\Delta_{\min} = 0.0137711 \text{ m}$. Applying a target Courant number of 0.5 gave

$$\Delta t_{\text{est}} = \frac{0.5\Delta_{\min}}{U_{\max}} = 5.85 \times 10^{-3} \text{ s}. \quad (1)$$

A slightly smaller fixed time step of $\Delta t = 0.005 \text{ s}$ was therefore selected.

Sampling and averaging

One nominal flow-through time was

$$T_{\text{FT}} = \frac{9H}{U_b} \simeq 9 \text{ s}. \quad (2)$$

The reported LES solution was advanced for 25 000 time steps, corresponding to 125 s. The first 9 000 time steps, or 45 s, were discarded before statistical sampling was activated. Approximately five flow-through times were therefore allowed for the initial transients to decay. The main statistical record covered approximately 80 s, corresponding to $8.9T_{\text{FT}}$. Fluent time-averaged flow statistics were accumulated every LES time step after $t = 45 \text{ s}$.

Instantaneous velocity histories used for the energy spectra and two-point correlations were also saved every time step, giving a sampling interval of $\Delta t_s = 0.005 \text{ s}$. These records were truncated at iteration 25 000 for consistency with the saved LES solution. Instantaneous wall-shear-stress data were saved every ten time steps, corresponding to $\Delta t_s = 0.05 \text{ s}$. The statistical homogeneity of the spanwise direction was exploited by extracting data at

$$z/H = 0.45, 1.35, 2.25, 3.15, 4.05. \quad (3)$$

For a general quantity ϕ , the reported time- and spanwise-averaged value was

$$\langle \overline{\phi} \rangle_z = \frac{1}{5} \sum_{j=1}^5 \left[\frac{1}{T} \int_{t_0}^{t_0+T} \phi(z_j, t) \text{ d}t \right]. \quad (4)$$

Model verification

Near-wall and bulk-flow resolution

The local friction velocity and wall-unit spacings were evaluated as

$$u_\tau = \sqrt{\frac{|\tau_{w,x}|}{\rho}}, \quad y^+ = \frac{\rho u_\tau \Delta y_1}{\mu}, \quad \Delta x^+ = \frac{\rho u_\tau \Delta x}{\mu}, \quad \Delta z^+ = \frac{\rho u_\tau \Delta z}{\mu}. \quad (5)$$

The lower-wall values of y^+ remained below approximately 0.3 over the complete hill geometry (Fig. 2a), well below the wall-resolved LES guideline of order unity [4]. On the flat valley floor, Δx^+ was approximately 10-50 and remained below the commonly used guideline of 100. The spanwise spacing was generally approximately 10-40, with local maxima of about 53 near the beginning of the flat valley floor. The wall-normal and streamwise resolutions were therefore satisfactory, although the approximate recommendation $\Delta z^+ \simeq 30$ was locally exceeded. The bulk-flow resolution was assessed using the precursor integral length scale

$$l_0 = \frac{\sqrt{k}}{0.09\omega}, \quad \Delta = V_{\text{cell}}^{1/3}. \quad (6)$$

The resulting l_0/Δ values were generally between approximately 8 and 27 through most of the assessed profiles (Fig. 2b). Lower values occurred near the walls, while the separated shear layer and bulk-flow regions were resolved by several cells per integral length scale. The mesh was therefore judged adequate for resolving the dominant energy-containing motions.

Temporal resolution

The temporal resolution was assessed using the local cell-convective Courant number. The centreplane distribution and selected wall-normal profiles remained below approximately 0.25, including the separated shear layer and near-wall regions (Fig. 3). The selected time step was therefore more conservative than the target $\text{CFL} \lesssim 1$.

SGS-model activity and two-point correlations

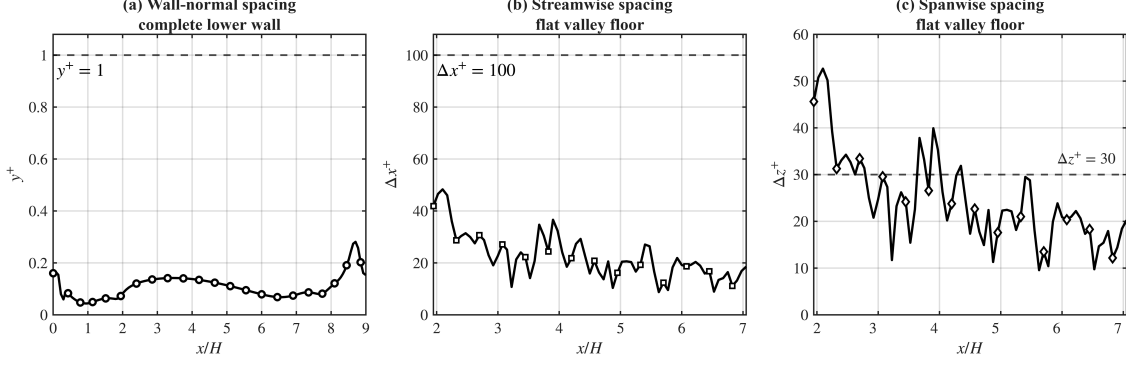
The activity of the WALE model was assessed from

$$\frac{\nu_{\text{SGS}}}{\nu} = \frac{\mu_{\text{SGS}}}{\mu}. \quad (7)$$

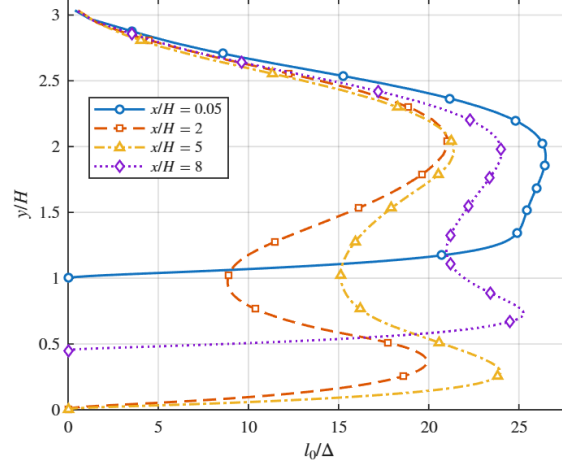
The time- and spanwise-averaged ratios at $x/H = 0.5$ and 6 remained below approximately 0.5 (Fig. 4a). The mean SGS viscosity was therefore smaller than the molecular viscosity at the assessed locations, indicating that the simulation was not dominated by excessive WALE viscosity. This ratio was treated as an indicator of subgrid-model activity rather than a complete measure of SGS dissipation [1]. The normalized two-point velocity correlation was defined as

$$R_{ii}(\mathbf{r}) = \frac{\overline{u'_i(\mathbf{x}, t) u'_i(\mathbf{x} + \mathbf{r}, t)}}{u_{i,\text{rms}}(\mathbf{x}) u_{i,\text{rms}}(\mathbf{x} + \mathbf{r})}. \quad (8)$$

The streamwise correlation used the u -velocity histories at 65 monitor points, while the spanwise correlation used the w -velocity histories at 56 points. The first and final monitors represented duplicate locations across the periodic boundaries, and one endpoint was therefore removed from each dataset. Monitor point 10 was used as the reference signal. The first zero



(a) Lower-wall y^+ , Δx^+ and Δz^+ .



(b) Wall-normal profiles of l_0/Δ .

Figure 2: Spatial-resolution verification: (a) near-wall grid spacing and (b) integral-length-scale resolution at selected streamwise stations.

crossing was obtained using linear interpolation between the last positive and first negative values.

The spanwise correlation decreased approximately monotonically and reached zero at $r_z/H \simeq 0.8$ (Fig. 4c). This distance corresponded to approximately ten spanwise computational-cell intervals, exceeding the approximate minimum of eight cells across a dominant resolved structure suggested for a coarse LES [4]. Since the five averaging planes were separated by $\Delta z/H = 0.9$, slightly greater than the zero-correlation distance, they were treated as approximately independent statistical samples.

The streamwise correlation approached zero only at approximately $r_x/H = 4.8$ and contained several secondary maxima. This non-monotonic behaviour was attributed to the spatially inhomogeneous nature of the periodic-hill flow, because the monitoring line crossed the separation, recirculation, reattachment and recovery regions. The streamwise result was therefore used as a qualitative measure of the large resolved structures rather than as a homogeneous integral length scale.

Kolmogorov-scale assessment

For the optional dissipative-scale assessment, the resolved dissipation and Kolmogorov length scale were estimated as

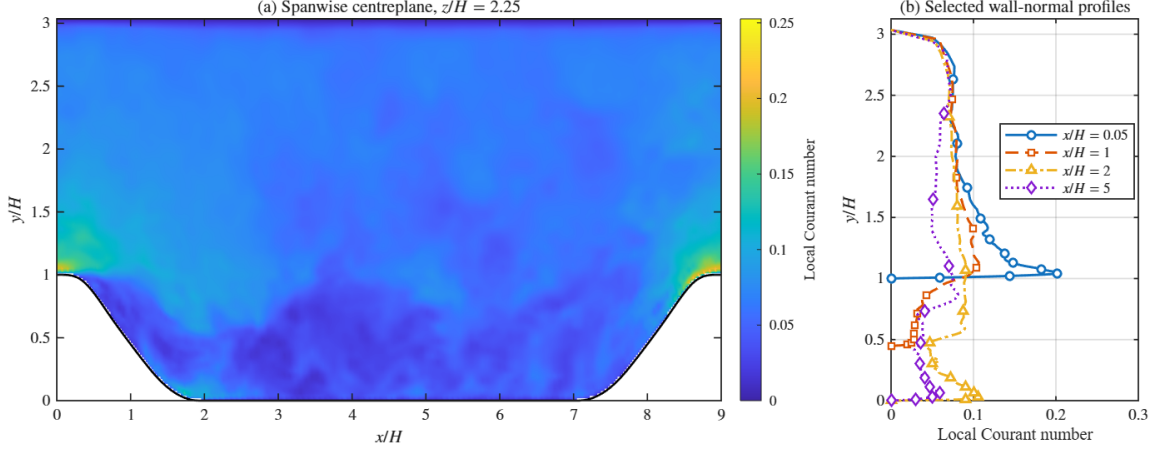


Figure 3: LES temporal-resolution assessment: (a) centreplane distribution of the local cell-convective Courant number and (b) profiles at selected streamwise stations.

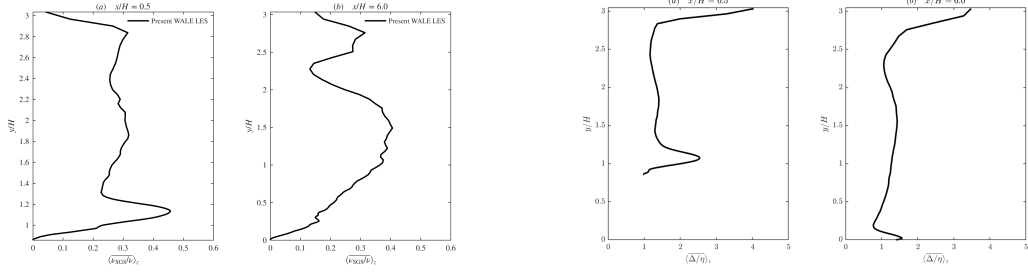
$$\varepsilon_{\text{res}} = 2\nu S_{ij}^f S_{ij}^f, \quad \eta = \left(\frac{\nu^3}{\varepsilon_{\text{res}}} \right)^{1/4}. \quad (9)$$

At $x/H = 0.5$ and 6 , the spanwise-averaged Δ/η values were generally between approximately 1 and 4 (Fig. 4b). The obtained values were below the $\Delta/\eta < 12$ level reported over most of the highly resolved LES of Fröhlich et al. [1] and were of the same order as the dissipative-scale resolution reported by Krank et al. [3]. The selected sections therefore resolved a substantial part of the dissipative range, although this conclusion was limited to the available profiles.

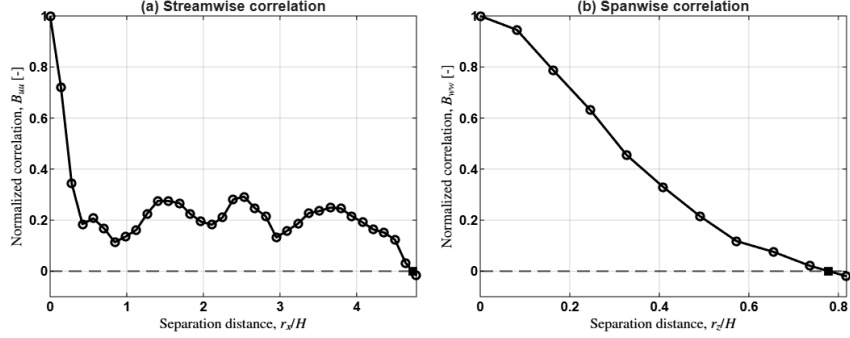
Statistical convergence

Statistical convergence was assessed over an approximately 80 s sampling interval, corresponding to $8.9T_{\text{FT}}$, after five flow-through times had been discarded. The record was divided into eight equal blocks, and cumulative estimates from six and eight blocks were compared. At the outer monitoring point, the mean streamwise velocity changed by only 4.25% when the final two blocks were included, indicating reasonable stability of the local first-order statistic.

The temporal autocorrelation of the instantaneous WSS quantities used to calculate TAWSS and transWSS was also examined. The corresponding integral time scale represents the approximate duration over which consecutive samples remain correlated. Median values of 5.78 s for TAWSS and 4.32 s for transWSS were obtained, with maximum values of 9.44 s and 7.57 s, respectively. The block comparison was therefore treated as a practical assessment of sensitivity to the available sampling duration rather than as a formal estimate based on statistically independent blocks. Despite the remaining temporal variability, averaging over five approximately independent spanwise locations produced consistent streamwise trends in TAWSS, transWSS and WSS directionality. The available record was therefore considered adequate for identifying broad regional differences along the valley floor, while individual pointwise values were interpreted as finite-sampling estimates.



(a) Time- and spanwise-averaged SGS-viscosity ratio at $x/H = 0.5$ and 6 . (b) Time- and spanwise-averaged Δ/η ratio at $x/H = 0.5$ and 6 .



(c) Streamwise and spanwise two-point velocity correlations.

Figure 4: Additional LES-verification measures: (a) WALE SGS-model activity, (b) optional dissipative-scale resolution assessment and (c) two-point correlations used to assess the dominant resolved structures and the independence of the spanwise averaging planes.

Validation and flow-structure assessment

The model-validation findings are presented in Section 3. Mean streamwise and wall-normal velocities, resolved Reynolds shear stress and resolved turbulent kinetic energy were extracted at

$$x/H = 0.05, 0.5, 1, 2, 3, 4, 5, 6, 7, 8 \quad (10)$$

and compared with the DNS data of Krank et al. [3]. Bottom-wall pressure and skin-friction coefficients were calculated as

$$C_p = \frac{p - p_{\text{ref}}}{\frac{1}{2}\rho U_b^2}, \quad C_f = \frac{\tau_{w,x}}{\frac{1}{2}\rho U_b^2}. \quad (11)$$

The precursor RANS, present LES and DNS skin-friction distributions were compared, and the mean separation and reattachment positions were obtained from the zero crossings of C_f using linear interpolation. Velocity spectra were calculated from the instantaneous signals at $(x/H = 2.15, y/H = 0.49)$ and $(x/H = 2.23, y/H = 1.13)$ representing the recirculation region and separated shear layer, respectively. The spectra of the three velocity components were averaged over the five spanwise positions in Eq. (3). Frequency and spectral density were normalised using H and U_b , and the spectral decay was compared with an $f^{-5/3}$ reference slope and the results of Fröhlich et al. [1]. Instantaneous coherent structures were visualised using a three-dimensional Q -criterion isosurface at $Q = 16.5 \text{ s}^{-2}$, coloured by velocity magnitude.

Wall-shear-stress assessment of the valley floor

The aerodynamic environment along the flat valley floor was assessed at

$$x/H = 2, 3, 4, 5, 6, 7 \quad (12)$$

and at each of the five spanwise locations in Eq. (3), giving 30 wall-monitor points. On the flat floor, the instantaneous wall-shear-stress vector was written as

$$\boldsymbol{\tau}_w(t) = \tau_{w,x}(t)\mathbf{e}_x + \tau_{w,z}(t)\mathbf{e}_z. \quad (13)$$

The time-averaged wall shear stress was calculated as

$$\text{TAWSS} = \frac{1}{T} \int_{t_0}^{t_0+T} |\boldsymbol{\tau}_w(t)| dt. \quad (14)$$

The mean WSS direction was defined by

$$\mathbf{e}_m = \frac{\overline{\boldsymbol{\tau}_w}}{|\overline{\boldsymbol{\tau}_w}|}, \quad (15)$$

and the direction transverse to the mean WSS by $\mathbf{e}_\perp = \mathbf{n} \times \mathbf{e}_m$, where $\mathbf{n}$ is the wall-normal unit vector. The transverse WSS was then calculated as

$$\text{transWSS} = \frac{1}{T} \int_{t_0}^{t_0+T} |\boldsymbol{\tau}_w(t) \cdot \mathbf{e}_\perp| dt, \quad (16)$$

following the directional formulation used by Andersson et al. [6]. The normalized ratio $\text{transWSS}/\text{TAWSS}$ was also evaluated. WSS wind roses were constructed from the instantaneous magnitude and direction

$$|\boldsymbol{\tau}_w| = \sqrt{\tau_{w,x}^2 + \tau_{w,z}^2} \quad (17)$$

$$\theta = \text{atan2}(\tau_{w,z}, \tau_{w,x}). \quad (18)$$

At each streamwise station, the samples from all five spanwise positions were combined without first averaging the WSS vectors. Identical WSS-magnitude bins and radial-frequency limits were used for the six wind roses to permit direct comparison. The candidate village region was assessed from the combined trends in mean velocity, C_f , TAWSS, transWSS, $\text{transWSS}/\text{TAWSS}$ and the directional stability of the instantaneous WSS.

3. Results

Validation of the mean flow and turbulence statistics

The mean streamwise velocity reproduced the principal development of the periodic-hill flow, including the velocity deficit downstream of the crest, the reversed near-wall flow within the valley, and the subsequent recovery towards the following hill (Fig. 5a). Agreement with the DNS was strongest in the outer flow and at the downstream recovery stations. The largest differences occurred below the separated shear layer, where the present LES predicted a shorter and weaker recirculation region than the DNS. The wall-normal velocity similarly reproduced

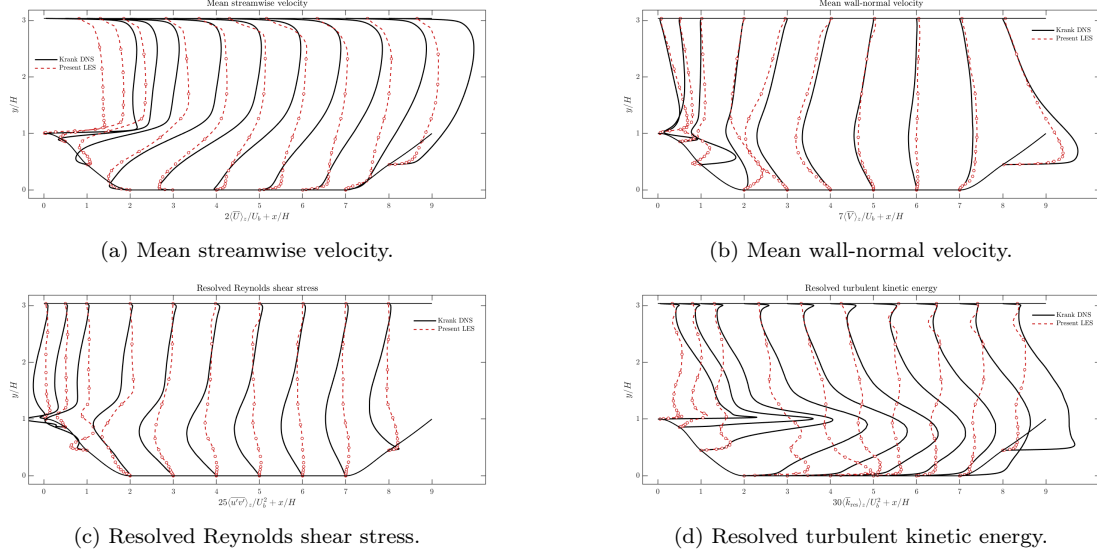


Figure 5: Comparison of the time- and five-plane spanwise-averaged LES profiles with the DNS data of Krank et al. [3]: (a) mean streamwise velocity, (b) mean wall-normal velocity, (c) resolved Reynolds shear stress and (d) resolved turbulent kinetic energy. Consecutive profiles are shifted in the streamwise direction using the scaling indicated on the corresponding horizontal axis.

the main signs and streamwise development of the vertical motion, but differences remained in the near-crest and early separated-flow regions (Fig. 5b).

The resolved Reynolds shear stress followed the expected concentration of turbulent momentum transport in the separated shear layer and its downstream decay (Fig. 5c). Its magnitude was nevertheless underpredicted at most stations, particularly within the shear layer above the recirculation zone. The resolved turbulent kinetic energy showed the same qualitative behaviour, with elevated values developing downstream of separation and decaying after reattachment (Fig. 5d). However, the LES values remained below the DNS reference over much of $1 \lesssim x/H \lesssim 6$. The comparison therefore indicates that the mean-flow development was represented more accurately than the second-order turbulent statistics. Similar TKE underprediction has also been observed for less highly resolved LES cases in the periodic-hill comparison of Krank et al. [3].

Wall coefficients and recirculation region

The bottom-wall skin-friction coefficient reproduced the main features of the separated periodic-hill flow, including negative values within the recirculation region, recovery after reattachment and the strong positive peak approaching the following hill (Fig. 6a). The principal DNS recirculation region extended from $x_{\text{sep}}/H = 0.1982$ to $x_{\text{reatt}}/H = 4.5112$, giving a bubble length of $L_b/H = 4.3130$. The precursor SST RANS predicted separation at $x_{\text{sep}}/H = 0.2709$, only $0.0727H$ downstream of the DNS location. However, negative skin friction persisted until $x_{\text{reatt}}/H = 7.6191$, giving $L_b/H = 7.3481$. The RANS model therefore overpredicted the principal recirculation length by 70.4%. The WALE LES predicted separation farther downstream at $x_{\text{sep}}/H = 1.2745$ and reattachment at $x_{\text{reatt}}/H = 3.9678$, giving $L_b/H = 2.6933$. Although the LES underpredicted the bubble length by 37.6%, its reattachment position and overall bubble length were considerably closer to the DNS than those of the SST RANS. The SST RANS therefore represented the initial separation position more accurately, whereas the LES gave the better prediction of reattachment and the total recirculation extent.

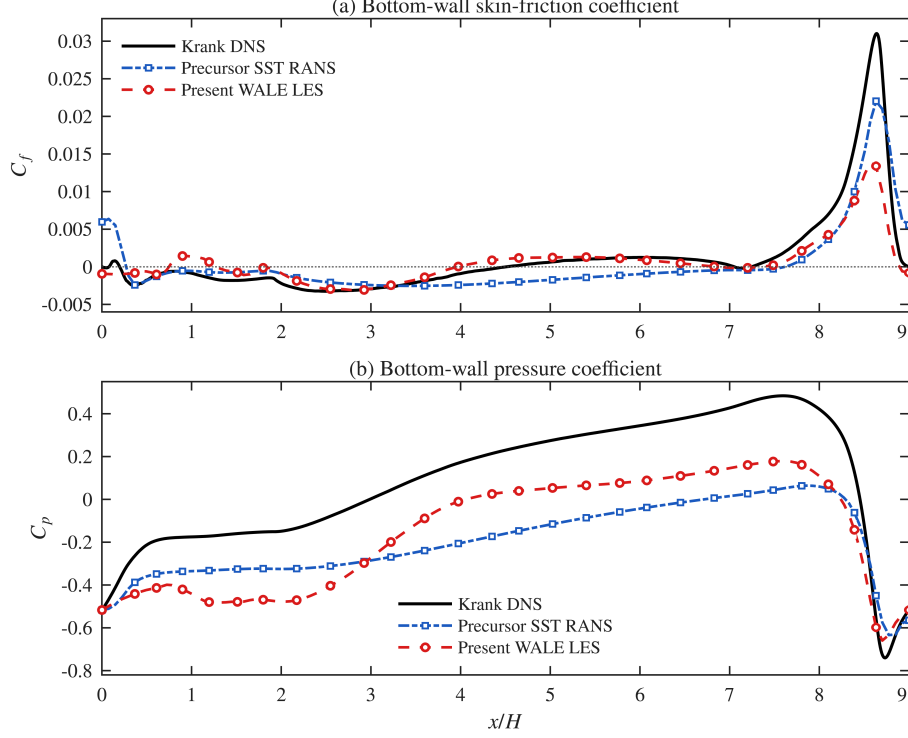


Figure 6: Comparison of the bottom-wall (a) skin-friction coefficient and (b) pressure coefficient obtained from the precursor SST RANS and present WALE LES with the DNS data of Krank et al. [3]. The RANS and LES quantities were averaged over the five spanwise locations. Since pressure in the periodic incompressible calculations contains an arbitrary additive reference, the numerical C_p curves were shifted to match the DNS value at $x/H = 0$.

Quantity	Krank DNS	SST RANS	WALE LES
x_{sep}/H	0.1982	0.2709	1.2745
x_{reatt}/H	4.5112	7.6191	3.9678
L_b/H	4.3130	7.3481	2.6933
Bubble-length difference	—	+70.4%	−37.6%

Table 2: Principal separation and reattachment locations determined from the negative- C_f intervals.

Both numerical models underpredicted the strong positive skin-friction peak near $x/H \simeq 8.6$. The SST RANS peak was closer to the DNS magnitude than the LES peak, while the LES more closely reproduced the DNS behaviour through the principal reattachment and downstream-recovery regions.

The pressure-coefficient distributions captured the overall pressure recovery across the valley and the rapid pressure reduction approaching the following crest (Fig. 6b). Both numerical models underpredicted the DNS pressure recovery. The SST RANS showed the weakest recovery over most of the valley, whereas the WALE LES produced a larger pressure increase and was generally closer to the DNS through the downstream half of the domain. These differences are consistent with the excessively long recirculation region predicted by the SST RANS and the shorter recirculation region predicted by the LES.

Resolved turbulent dynamics

The velocity spectra at the recirculation and separated-shear-layer monitoring points contained broadband energy rather than a single dominant frequency (Fig. 7a). At both locations, the energy decreased progressively with increasing frequency, and part of the intermediate-frequency range followed a trend approximately parallel to the $f^{-5/3}$ reference slope. The three component spectra also approached one another at the higher resolved frequencies, consistent with increasing small-scale isotropy.

Compared with the highly resolved LES of Fröhlich et al. [1], the high-frequency roll-off occurred earlier, particularly at the separated-shear-layer point. This indicates stronger effective dissipation arising from numerical and/or SGS contributions in the present calculation. Nevertheless, the absence of a high-frequency energy accumulation and the presence of a limited inertial-type decay demonstrate that a range of turbulent scales was resolved. The instantaneous Q -criterion field contained numerous three-dimensional vortical structures within the separated shear layer and recirculation region (Fig. 7b). Structures close to the valley floor were predominantly associated with lower velocity magnitudes, whereas more elongated and energetic structures occurred near the hill crests and along the accelerating windward flow. The distribution confirms that the LES captured the shedding, convection and breakdown of coherent structures, although the visualization does not by itself quantify their occurrence frequency.

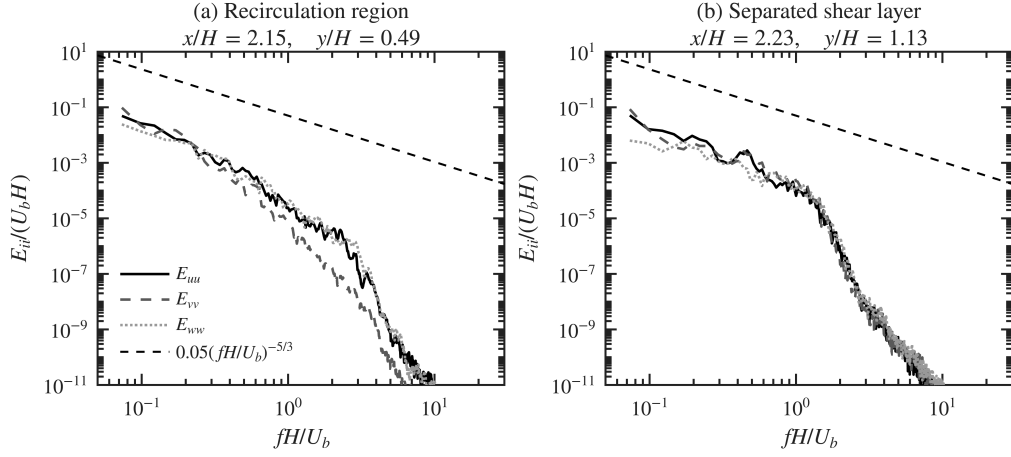
Aerodynamic environment along the valley floor

The spanwise-averaged TAWSS initially increased from approximately 2.5×10^{-4} Pa at $x/H = 2$ to a maximum of about 4.6×10^{-4} Pa at $x/H = 3$ (Fig. 8a). It then decreased progressively through the downstream half of the flat valley, reaching approximately 1.4×10^{-4} Pa at $x/H = 6$ and 0.5×10^{-4} Pa at $x/H = 7$. Thus, the lowest mean near-ground loading occurred towards the downstream end of the flat floor. The largest absolute transverse WSS occurred at $x/H = 4$, where transWSS was approximately 1.9×10^{-4} Pa (Fig. 8b). The normalized value also reached its maximum at this station,

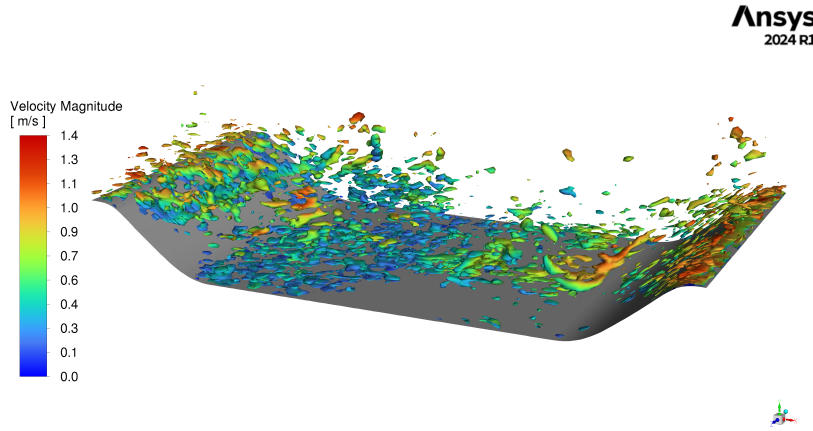
$$\frac{\text{transWSS}}{\text{TAWSS}} \simeq 0.61,$$

indicating that transverse disturbances were particularly important close to the mean reattachment region (Fig. 8c). By contrast, both the absolute and normalized transverse measures were low at $x/H = 5$ and 6. Although the absolute transWSS was lowest at $x/H = 7$, the normalized ratio increased again to approximately 0.25, and the local mean streamwise shear reversed direction. The spanwise ranges were also largest around $x/H = 4$, indicating greater spatial variability near reattachment.

The WSS wind roses provide the full directional context for these scalar measures (Fig. 9). At $x/H = 2$, substantial occurrence was observed in both the $+x$ and $-x$ directions, consistent with an intermittently reversing near-wall flow. The distribution at $x/H = 3$ was dominated by the upstream-oriented $-x$ direction and included the largest WSS-magnitude classes. At $x/H = 4$, the dominant direction had changed to $+x$, but the elevated transWSS showed that transverse disturbances remained important near reattachment. The distributions at $x/H = 5$ and 6 were narrow and strongly concentrated in the downstream $+x$ direction. This indicates a more predictable near-wall environment with limited directional variability. At $x/H = 7$, the dominant direction returned to $-x$, consistent with the secondary negative- C_f region before the



(a) Velocity-component energy spectra at the recirculation and separated-shear-layer monitoring points.



(b) Instantaneous Q -criterion isosurface at $Q = 16.5 \text{ s}^{-2}$, coloured by velocity magnitude.

Figure 7: Resolved turbulent dynamics: (a) turbulence energy spectra compared with an $f^{-5/3}$ reference slope and (b) instantaneous coherent vortical structures.

following hill. The predominantly empty $\pm z$ sectors at all stations show that the WSS remained mainly aligned with the streamwise axis, while the transverse measure distinguished the smaller directional fluctuations that were not evident from C_f or TAWSS alone.

Considering the combined criteria of low mean loading, low transverse disturbance and stable direction, the region around

$$5.5 \lesssim x/H \lesssim 6.5 \quad (19)$$

provided the most favourable location for the proposed village. This region combined relatively low TAWSS and transWSS with a narrow, consistently downstream-oriented WSS distribution. The station at $x/H = 7$ had a lower absolute WSS magnitude, but its reversed mean direction and proximity to the secondary separation region made it a less robust choice. The reattachment region around $x/H = 4$ and the upstream recirculation region around $x/H = 2-3$ were unsuitable because of their stronger directional variability, flow reversal and, at $x/H = 3$, comparatively high mean wall loading.

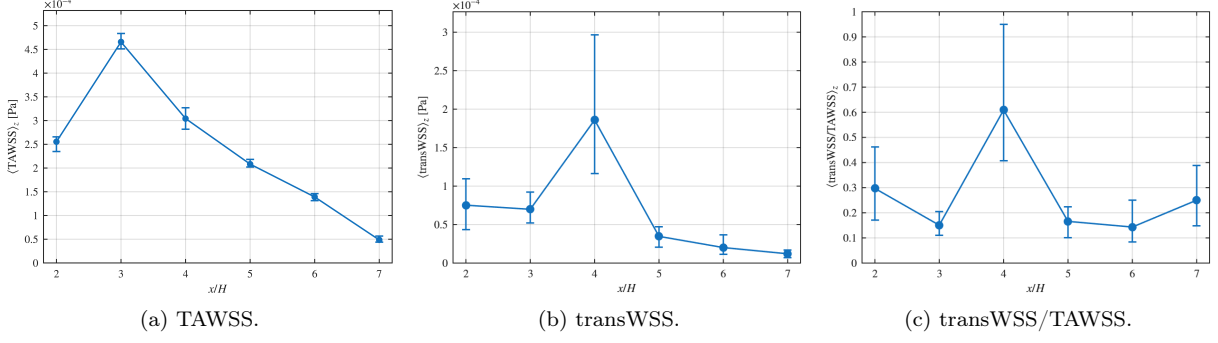


Figure 8: Wall-shear-stress measures along the flat valley floor: (a) time-averaged WSS magnitude, (b) transverse WSS and (c) transverse WSS normalized by TAWSS. Markers denote averages over the five spanwise locations, while error bars indicate their minimum and maximum values.

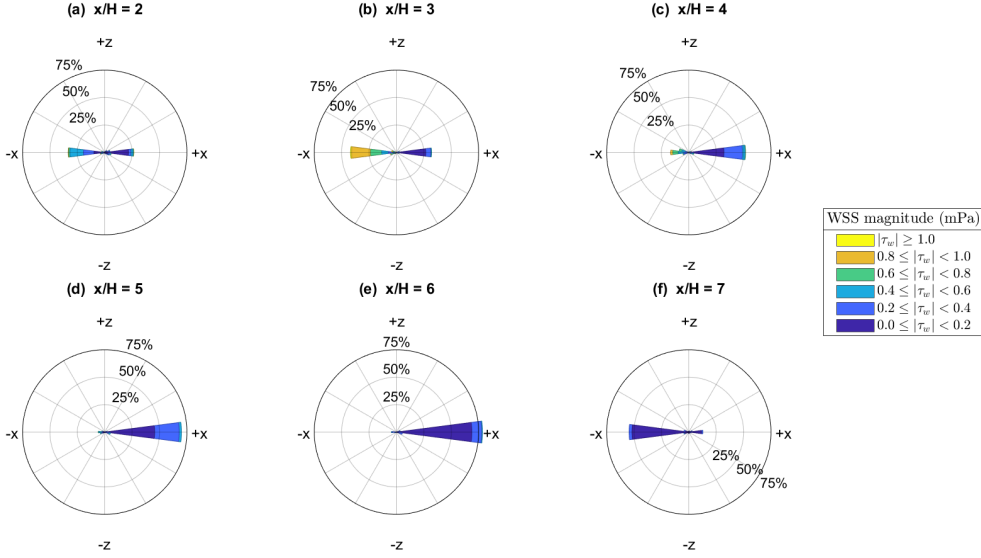


Figure 9: Directional frequency distributions of the instantaneous wall-shear-stress vector at six streamwise stations along the flat valley floor. Samples from $z/H = 0.45, 1.35, 2.25, 3.15$ and 4.05 were combined at each station. The radial coordinate denotes occurrence frequency and colour denotes WSS magnitude. Common magnitude and frequency scales are used in all panels.

4. Discussion

The verification results indicate that the selected grid and time step were adequate for resolving the principal separated-flow structures. The low y^+ , acceptable streamwise wall spacing, CFL values below unity, moderate SGS-viscosity ratio and two-point correlations support the use of the present LES for studying the overall flow development. The validation nevertheless showed that the accuracy was greater for the first-order mean flow than for the second-order turbulence statistics. The LES captured the velocity deficit, reversed flow, separated shear layer and downstream recovery, but predicted separation too far downstream and reattachment too far upstream, resulting in a shorter recirculation region than the DNS of Krank et al. [3]. The resolved TKE and Reynolds shear stress were also underpredicted over much of the valley. These discrepancies are consistent with the sensitivity of curved-surface separation to near-wall resolution and the greater numerical and SGS dissipation of a coarser LES than the highly resolved reference calculations [1, 3]. The shorter averaging interval also contributed uncertainty,

particularly for the slowly varying separated-flow and WSS statistics. The pressure-coefficient comparison should additionally be interpreted cautiously because its absolute level is sensitive to the selected reference pressure.

The comparison with the precursor SST RANS also illustrates the different behaviour of the two turbulence-modelling approaches. The steady RANS model represents the full turbulent field through modelled transport equations, whereas LES resolves the unsteady energy-containing structures and models only the smaller subgrid scales. The SST RANS predicted the initial separation position more accurately, but delayed reattachment to $x/H = 7.6191$ and overpredicted the principal bubble length by 70.4%. The LES predicted separation too far downstream, but gave a substantially more accurate reattachment position and reduced the bubble-length error to 37.6%. In addition, only the LES supplied the instantaneous information required for the energy spectra, coherent-structure visualisation and directional WSS assessment.

No individual WSS parameter fully describes the aerodynamic environment. The signed skin-friction coefficient identifies mean flow reversal, separation and reattachment, but cannot represent temporal or spanwise directional fluctuations. TAWSS describes the mean magnitude of the near-ground loading, but removes all directional information; a low value may therefore occur during either consistently weak flow or repeated opposing shear. TransWSS complements TAWSS by quantifying disturbances perpendicular to the mean WSS direction and was particularly effective in identifying the multidirectional environment near reattachment. However, it does not directly capture purely streamwise reversals and becomes more sensitive when the mean WSS vector is small. Similarly, transWSS/TAWSS facilitates comparison between locations, but may appear large where TAWSS is very low. Wind roses retain the joint frequency, direction and magnitude of the instantaneous WSS and therefore reveal reversal and directional stability directly, although they are less convenient as a single quantitative ranking measure. The combined application of these complementary measures is therefore more reliable than using C_f or TAWSS alone, consistent with the broader directional information attributed to transWSS in complex flows by Andersson et al. [6]. Based on the two practical criteria of low loading and predictable direction, the region

$$5.5 \lesssim x/H \lesssim 6.5$$

is the preferred location for the tourist village. This part of the valley combined relatively low TAWSS and transWSS with a narrow WSS distribution predominantly aligned in the downstream direction. The region near $x/H = 4$ was less suitable because reattachment produced the largest transverse and spanwise variations, while $x/H = 2-3$ experienced recirculation, reversal and stronger directional variability. Although $x/H = 7$ had the lowest absolute TAWSS, its predominantly reversed WSS and proximity to the secondary separation zone make it less predictable than $x/H = 5-6$.

Confidence in the exact numerical WSS values is limited by the approximately 8.9 flow-through-time averaging interval, since the block analysis showed greater sampling sensitivity for TAWSS and transWSS than for the local mean velocity. The recommendation should therefore be interpreted as a favourable *region*, rather than a precise optimum position. Its confidence is nevertheless strengthened by the consistent ranking obtained from TAWSS, transWSS, the normalized transverse measure, C_f and the wind roses. Additional uncertainty arises from representing real terrain by smooth periodic hills, assuming one persistent wind direction, neglecting atmospheric stability and surface roughness, and excluding the influence of buildings. Future work should employ longer sampling, additional wall-monitor locations, further mesh refinement

and several incident wind directions, followed by simulations including the local terrain and proposed structures.

5. Conclusions

A wall-resolved WALE LES was used to investigate flow over periodic hills at $Re_H = 10\,595$ and to identify a suitable location for a tourist village. The verification measures showed adequate spatial and temporal resolution for resolving the principal turbulent structures. The LES captured the main mean-flow behaviour and three-dimensional vortex dynamics, although the recirculation region was shorter and the resolved TKE and Reynolds shear stress were lower than the DNS reference values. Compared with the precursor SST RANS, the WALE LES gave a substantially better prediction of the reattachment position and overall recirculation extent, although its initial separation position remained too far downstream.

The combined WSS assessment showed that mean loading alone was insufficient for judging the local environment. TAWSS quantified its magnitude, transWSS identified multidirectional disturbances, and the wind roses distinguished stable downstream flow from intermittent or reversed shear. Together, these measures identify $5.5 \lesssim x/H \lesssim 6.5$ as the most favourable valley region because it provides relatively low WSS together with the most consistent downstream-oriented direction. The recommendation is considered robust as a regional trend, but its precise boundaries remain subject to statistical and modelling uncertainty and should be confirmed using longer sampling and a more site-specific terrain model.

Contribution Statement

The authors jointly contributed to the planning and execution of this study. Neha Mandapaka and Mukundan Subbaiya contributed to the simulation setup, data extraction, post-processing, interpretation of the results and preparation of the report. Both authors reviewed the final report, approved the submitted version and take responsibility for the accuracy and integrity of the work.

Use of Generative AI and AI-Assisted Technologies

During the preparation of this report, the authors used ChatGPT to assist with MATLAB-code development, post-processing workflows, organisation of the report, interpretation of numerical results and language revision. All generated material was critically reviewed, verified and edited by the authors. The authors take full responsibility for the methods, calculations, interpretations and final content presented in this report.

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